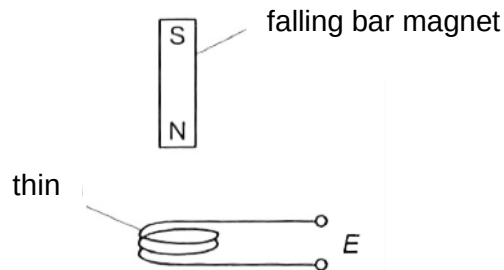
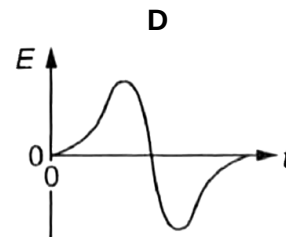
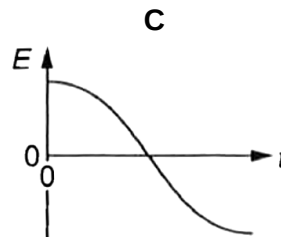
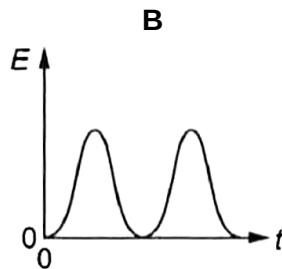
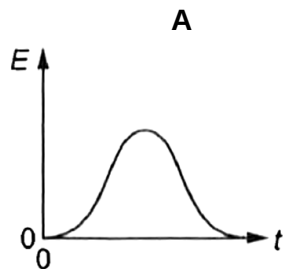


net B at point O . $\therefore$ i.e. current in all four wires must be in the same direction.

- 25** In an experiment to record electrical events of short duration, a student drops a bar magnet through a very thin, horizontal coil, as shown.



Which graph best represents how the induced e.m.f. E in the coil varies with time t ?



Ans: D

When the north pole of the magnet approaches the coil, there is an increase in flux linkage. Hence an emf and current is induced in a direction to produce a B field pointing upward to oppose the increase in flux linkage. When the magnet leaves the coil, an emf and current is now induced in a direction to produce a B field pointing upward to oppose the decrease in flux linkage. Answer has to be D because the induced emf, which is the rate of change of flux linkage should start from zero.

26. A uniform magnetic field directed at 50° from the vertical passes through a circular metal ring of